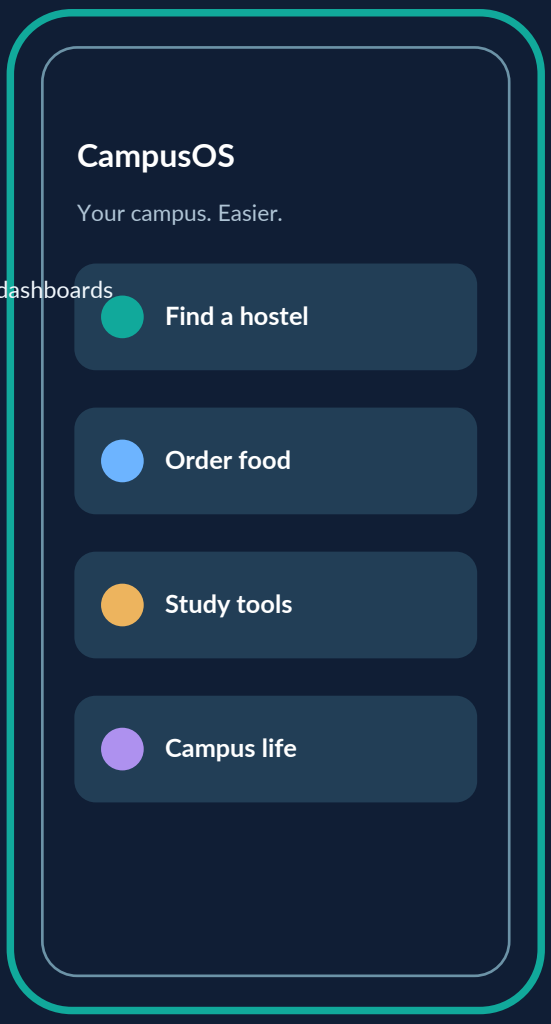
THE COMPLETE
PRODUCT BLUEPRINT

CampusOS Nigeria

A housing-first student super app for academic life, campus commerce, and everyday student needs.

Concept • Student mobile app • Discovery website • Landlord & vendor dashboards



PRODUCT STRATEGY / DESIGN / DELIVERY

NOVEMBER 2026 • WORKING BLUEPRINT • VERSION 1.0

EXECUTIVE SUMMARY

Build a trusted campus ecosystem - one school at a time

CampusOS is the working name for a Nigerian student platform serving universities, polytechnics and colleges of education. The product starts with a credible housing marketplace, then builds daily usefulness through food, services and practical academic tools.

> **The central product thesis**
> Housing can attract students and provide a monetization path through landlords and authorized agents. Academic tools create repeat engagement. Food, shopping and campus services can build transaction frequency - only after local supply and service quality are reliable.

1 campus
Start narrow and build supply density

3 pillars
Housing, campus essentials, study tools

Free search
Keep core student discovery accessible

What we are building

- **Student mobile app:** find verified housing, compare total costs, request inspections, discover vendors, and organize school life.
- **Public website:** search-friendly pages for schools, campus areas, housing and campus businesses.
- **Business portal:** landlord, agent and vendor tools for listings, availability, enquiries, orders and reporting.
- **Operations console:** verification, moderation, disputes, support, payments, announcements and analytics.

Non-negotiable principles

PRINCIPLE	PRODUCT DECISION
Trust over volume	A smaller set of checked, current listings is better than thousands of stale or fraudulent records.
Campus first	Every result is scoped to the student's institution, specific campus, and nearby neighborhoods.
Student affordability	Free essential search, complete fee disclosures, transparent vendor prices, low-data design.
Operations before automation	Human review where mistakes could cost students money or compromise safety.

Status: This is a product planning document, not a claim that the app is built, institutions are partners, rooms are available, or pricing has been validated. Figures and schedules marked proposed or illustrative must be tested.

NAVIGATION

Contents

The report separates the long-term platform vision from a realistic first launch. The numbered chapters can be used as a product and implementation reference.

Vision, positioning and scope 4
The problems and the people 5
The connected product ecosystem 6
Housing: the flagship marketplace 7
Food, groceries and everyday buying 8
Marketplace and campus services 9
Academics and study tools 10
Campus life, news and opportunities 11
User roles and end-to-end journeys 11
App and website information architecture 13
Monetization and financial hypotheses 14
Launch strategy: one campus, high reliability 15
Technical architecture 16
Data model and API boundaries 17
Trust, safety, privacy and compliance 18
Operations, service quality and human workflows 19
Phased delivery roadmap 20
MVP requirements and acceptance criteria 21
Success metrics and experiment plan 22
Risk register and decision log 22
The first 30 actions 23
References, assumptions and ownership 25

How to use this document

Start with Chapters 01-04 for scope and user value; use Chapters 09-14 as a development brief; use Chapters 15-20 for launch, operations and decision-making.

CHAPTER 01 / PRODUCT BLUEPRINT

Vision, positioning and scope

A student lifestyle platform with housing as the initial commercial wedge.

Vision statement

Make it easier for students in Nigeria to **live, learn, buy, connect and earn** within their campus ecosystem - starting with the hardest everyday friction: finding a safe, suitable place to stay.

Positioning

OPTION	UPSIDE	TRADE-OFF
Student super app	Clear long-term promise; easy to add study, food and services.	Too broad for early marketing; empty sections could reduce trust.
Housing-first student brand	Focused acquisition message; high-intent searches; clear landlord value.	Needs careful brand design to expand beyond accommodation.

Recommended positioning

Externally lead with **trusted housing near your school**. Inside the product, create a broader campus home. Position the brand as student-first from day one, and introduce additional modules when they can deliver real value.

Target institutions

- Universities: federal, state and private; support multiple campuses within an institution.
- Polytechnics: ND and HND course structures, semesters and departmental contexts.
- Colleges of education and other relevant tertiary institutions: configurable academic structures.

In scope vs not yet

BUILD EARLY	DEFER UNTIL VALIDATED
Housing listings, filters, availability, enquiry, inspection booking	In-app escrow, rent financing, lease signing, automatic deposits
Basic timetable, reminders, campus directory	AI tutor, automatic school portal integration, grade scraping
Vendor discovery and simple pickup pre-orders	Owned delivery fleet, nationwide logistics, complex subscriptions
Owner/agent verification, reports and moderation	Automated verification without human audit

CHAPTER 02 / PRODUCT BLUEPRINT

The problems and the people

Key pain points translated into personas and jobs to be done.

Problem map

NEED	CURRENT FRICTION	PROPOSED RESPONSE
Accommodation	Hidden charges, stale availability, difficult inspections, weak trust.	Verified listings, fee breakdown, map search, inspection requests.
Meals and supplies	Limited visibility of nearby prices and availability.	Campus food directory, menu visibility, pickup and basic ordering.
Campus transactions	Fragmented WhatsApp posts, unknown sellers and service providers.	Scoped listings, clear profiles and moderation.
Academics	Timetables and deadlines scattered across messages.	Simple planner, CGPA calculator, course resources.
Campus information	Rumors mixed with official announcements.	Source-labeled notices, event calendars and clubs.
Income	Student skills do not easily reach paying customers.	Student storefronts, tutoring and verified opportunities.

Primary personas

PERSONA	MAIN GOAL	SUCCESS SIGNAL
Newly admitted student	Understand the campus and find a room without being exploited.	Books a real inspection with transparent pricing.
Returning student	Renew, change room, buy essentials, track deadlines.	Compares options and returns weekly to useful tools.
Landlord / authorized agent	Fill genuine vacancies with suitable tenants.	Receives qualified enquiries and updates availability.
Food / campus vendor	Reach verified nearby demand and manage orders.	Gets repeat pickup orders or enquiries.
Admin / campus operator	Keep the directory safe, current and trustworthy.	Resolves reports and outdated listings promptly.

Validation interviews

Before building, interview approximately **30 students, 10 property providers and 10 campus vendors**. These are research targets, not evidence that demand already exists. Ask about the last real transaction, what failed, actual prices paid, workarounds, trust concerns and willingness to pay.

> **Design for mobile realities**
> Optimize for Android-first usage, patchy connections, low storage, compressed images, clear naira pricing and link sharing through WhatsApp. Avoid requiring continuous network access for basic academic information.

CHAPTER 03 / PRODUCT BLUEPRINT

The connected product ecosystem

Six modules built on one campus profile, shared trust and a shared search experience.

MODULE	STUDENT VALUE	STAGE
01 · Housing	Discover accommodation, costs, inspections and roommates.	MVP
02 · Food & groceries	Find meals, order pickup, discover daily essentials.	Phase 2
03 · Marketplace & services	Buy/sell used items; print, laundry, grooming, repairs.	Phase 2
04 · Academic life	Timetables, reminders, CGPA, materials, study help.	MVP basic; later deep
05 · Campus life	Events, verified notices, club groups, campus map.	Phase 3
06 · Opportunities	Tutoring, gigs, internships, scholarship links.	Phase 3-4

Cross-cutting services

- Account and campus identity: choose institution, campus and optional department and program.
- Messaging and notifications: inspection updates, academic deadlines, order status and moderation alerts.
- Trust services: listing verification, identity checks as appropriate, report/block, audit logs.
- Search and discovery: campus scoping, pricing, categories, saved searches and relevant recommendations.
- Payments and analytics: introduce only where a sound transaction and support model exists.

Network effects we want - in sequence

1. Supply — Real rooms listed

2. Trust — Rooms verified

3. Demand — Students search

4. Repeat — Study + commerce

An empty food feed or inaccurate hostel listing is worse than a smaller, reliable product. Add each new module only when there is enough supply to produce a satisfying experience.

CHAPTER 04 / PRODUCT BLUEPRINT

Housing: the flagship marketplace

The clearest early problem, strongest need for verification, and first revenue hypothesis.

Core student functionality

FEATURE	FUNCTIONAL DETAIL	PRIORITY
School and location filters	Institution, campus, nearby neighborhoods, distance estimate or map pin.	P0
Housing types	Room, self-contained, shared apartment, private hostel bed space.	P0
Budget comparison	Annual/session rent plus agent, agreement, caution and utilities fees.	P0
Listing media	Compressed photo gallery; video tour link or upload when practical.	P0
Availability status	Available, reserved, occupied, paused; last-confirmed date.	P0
Inspection booking	Request a time; provider accepts/declines; track status.	P0
Save and alerts	Favorites and optional notifications for matching new rooms.	P1
Roommate matching	Share preferences; introductions with safety controls.	P2
Reviews and issues	Post-tenancy reviews; disputes and evidence-based moderation.	P2

What every property listing must show

- Property identity: school/campus association, area, approximate map location and provider profile.
- Room inventory: number of available units or bed spaces, unit type, capacity, furnishing and restrictions.
- Amenities: electricity experience, water source, bathrooms, cooking facilities, internet and security measures.
- Complete price breakdown: base rent, agent charge, agreement, caution/deposit, service charges and payment period.
- Evidence: recent photos/video, date last checked, what was actually verified and who checked it.
- Actions: request inspection, ask a question, save property, report suspected fraud or a misleading listing.

Trust and verification ladder

STATUS	WHAT WE CHECK	UI LABEL
Submitted	Basic data is present; not yet screened.	Unverified · limited exposure
Identity checked	Landlord/agent identity and contact ownership reviewed.	Identity checked
Authority checked	Reasonable evidence of ownership or authorization to market.	Provider checked
Property inspected	Location and advertised condition reviewed by staff/approved verifier.	Property inspected · date

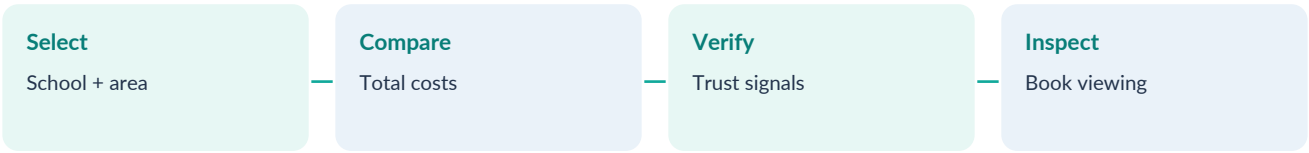
Critical rule for official school hostels

Do not facilitate unauthorized resale or transfers of school-allocated bed spaces. Official university hostel information should link to authorized school processes unless an institution formally partners with the platform.

Landlord and agent dashboard

- Create organization profile and submit verification documents through restricted access.
- Add properties, units, amenities, costs, media and available inventory.
- Approve and reschedule inspection requests; answer student enquiries.
- See listing views, saves, enquiries and conversion to physical inspections.
- Receive expiry reminders and reconfirm availability; pause a property immediately.

Housing workflows



For launch, rental negotiation and payment may occur outside the platform with transparent warnings. Avoid advertising "escrow" or deposit protection unless a suitable regulated arrangement truly exists.

Suggested housing matching score

Start with explicit filters rather than expensive AI. Example optional ranking: **35% budget fit + 25% proximity + 20% amenities + 15% verification / freshness + 5% user preferences**. Hide unavailable units; do not promote sponsored rooms as objectively best matches. Weights require user testing.

CHAPTER 05 / PRODUCT BLUEPRINT

Food, groceries and everyday buying

A potential daily-return loop - but only after local vendor operations are reliable.

Student functions

- Browse restaurants, bukas, snacks, bakeries and campus groceries scoped to school and delivery/pickup zones.
- View current menus, meal photos, prices, preparation windows and whether a vendor is accepting orders.
- Filter affordable meals by total cost; show packaging, pickup and delivery charges before checkout.
- Pre-order for pickup, or request delivery where an actual delivery process exists.
- Place group orders with clear payment responsibility and cancellation rules.
- Follow favorite vendors and see verified business discounts or limited promotions.

Vendor functions and order states

VENDOR CONSOLE	EXPECTED BEHAVIOR
Store and catalog	Manage hours, items, prices, stock, images and campus service area.
Orders	Accept, decline, mark preparing, ready, collected, completed or cancelled.
Reliability	Show realistic prep time, temporary closure and out-of-stock indicators.
Operations	Track completed orders, refund requests, support issues and payout records.

Rollout decision

Phase 2 should begin as **food discovery and pickup pre-order** with a small selection of vendors. Add courier delivery only after testing demand density, rider availability, support and economics.

Proposed transaction rules

- Confirm vendor acceptance before treating an order as fulfilled; track timeouts and status changes.
- Use server-verified payment webhooks if in-app checkout is enabled; never trust only the client success screen.
- Define when students can cancel, who pays delivery fees, and which failures trigger refunds.
- Do not claim food hygiene certification unless a legitimate verification process supports the claim.

CHAPTER 06 / PRODUCT BLUEPRINT

Marketplace and campus services

A practical discovery engine for student-to-student resale and nearby service providers.

CATEGORY	EXAMPLES	INITIAL EXPERIENCE
Used goods	Books, laptops, calculators, fans, mattresses, furniture.	Post item, condition, price, school and photos; contact seller.
Print and copy	Printouts, binding, photocopying, document scanning.	Price list, upload/request quote, pickup slot.
Laundry and cleaning	Wash/fold, ironing, room cleaning.	Provider profile, service menu, booking enquiry.
Beauty and care	Barber, hair, tailoring, nails.	Profiles, portfolios, prices, appointment enquiry.
Repairs and errands	Phone/laptop repairs, household maintenance, courier errands.	Provider ratings and service area; request quote.
Move-in/out	Secondhand room essentials and transport.	Seasonal marketplace collection and saved search.

Minimum trust controls

- Item listings expire or require reconfirmation; remove sold goods promptly.

- Mark commercial sellers separately from occasional student sellers.
- Provide report, block and dispute channels; monitor fake payments and impersonation.
- Discourage advance transfers to strangers, especially for high-value items.
- Restrict illegal, counterfeit, prohibited or unsafe goods and services.

Why this matters commercially

Campus services are a lower-frequency but broad supply opportunity. Business storefront upgrades, verified placement and lead packages may monetize better than charging every student seller. The first version can use enquiry links and scheduled pickup rather than a full escrow marketplace.

CHAPTER 07 / PRODUCT BLUEPRINT

Academics and study tools

Make the app valuable even when students are not spending money.

Launch essentials

TOOL	BEHAVIOR	NOTE
Lecture timetable	Weekly schedule, class/course, room/venue and reminders.	Offline-friendly local cache.
Assignment tracker	Manual due dates, priority, course and notifications.	No auto-submit promises.
Exam planner	Exam dates, revision blocks, progress checklist.	Student-controlled data.
GPA / CGPA calculator	Configurable grade bands, credit units, semesters.	Institution-specific rules; not an official transcript.
Academic calendar	Important dates with source and last updated stamp.	Separate official from community-supplied.

Later academic enhancements

- Course-specific group spaces with moderated Q&A;, class representatives and document collections.
- Authorized past questions and student-written notes with permissions and takedown processes.
- AI study assistant for explaining concepts, generating practice questions and drafting personal study plans.
- PDF/timetable extraction when budget allows, with user confirmation before saving extracted dates.
- Scholarship, internship and student loan information that links to verified official sources.

> **Academic integrity and accuracy**
> Do not offer to impersonate students, access restricted school accounts, or complete graded work on their behalf. Clearly label grade estimates and unofficial calendars. Prefer official institutional information where available.

CHAPTER 08 / PRODUCT BLUEPRINT

Campus life, news and opportunities

Community utility without turning the app into an unmoderated rumor feed.

Campus life

- A school and department-aware feed with content source labels: **official**, **verified group**, or **community**.
- Event calendars for orientation, clubs, department programs, sports, workshops and approved ticketed events.
- Campus maps and directory for faculties, libraries, health centers, offices and essential services.
- Lost-and-found reporting with safe contact flows and sensitive-information removal.
- Student union and club pages with verified ownership and moderation tools.

Student opportunities

OPPORTUNITY	STUDENT ACTION	SAFETY RULE
Tutoring	List subjects and rates, book a session.	No ghostwriting or exam cheating.
Student services	Create skills storefront and portfolio.	Moderate false claims and prohibited work.
Part-time gigs	Browse paid local tasks and internships.	Verify employers and flag exploitative offers.
Scholarships / loans	Find curated applications and official links.	Never request loan-portal passwords.
Campus ambassadors	Apply for approved community roles.	Transparent incentives and responsibilities.

Financial aid links should lead users directly to official institutions and providers. For student loans, use the official NELFUND resources rather than implying that CampusOS approves or disburses loans.

CHAPTER 09 / PRODUCT BLUEPRINT

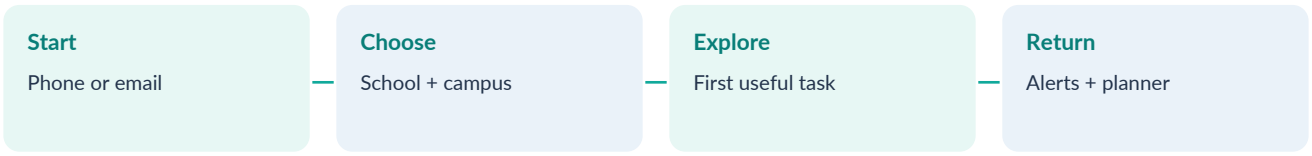
User roles and end-to-end journeys

Clear permissions, onboarding and task flows for every audience.

Account roles

ROLE	CAN DO	CANNOT DO
Visitor	Browse public school listings and guides.	View private messages or protected verification files.
Student	Save rooms, send requests, plan studies, create allowed listings.	Publish a verified badge or modify another student's data.
Property provider	Manage own properties, units and inspection requests.	Self-approve verification or access other providers' enquiries.
Vendor	Manage own store, orders, services and availability.	Alter payment ledger or impersonate admins.
Moderator / verifier	Review assigned content and evidence.	Perform sensitive actions without audit.
Platform admin	Manage institutions, roles, escalations and policy.	Bypass auditing or disclosure controls.

Student onboarding



First-session experience

- Allow browsing before account creation wherever possible.
- Prompt for school and campus, but explain why each detail is requested.
- Ask whether the student wants accommodation, services or academic planning first.
- Request notification permission after demonstrating value, not immediately on first screen.
- Use progressive profiles: department/year can be optional unless a specific feature requires it.

Operational journeys

- **Landlord:** register → submit identity/authorization → verification → publish room → reconfirm availability → reply to inspection → mark occupied.
- **Vendor:** register store → confirm campus zone → post menu → enable pickup → accept order → close order → reconcile payout.
- **Admin:** review queue → document evidence → approve/reject/request changes → monitor incidents → handle appeals.

CHAPTER 10 / PRODUCT BLUEPRINT

App and website information architecture

What users will actually see and navigate.

Recommended mobile navigation

TAB	PRIMARY CONTENT	SECONDARY SCREENS
Home	School-scoped search, shortcuts, timely cards, saved status.	Personalized suggestions, urgent announcements.
Explore	Housing, food, shop, campus services, filters.	Listing details, saved searches, location map.
School	Timetable, tasks, GPA, calendar and course materials.	Course pages, study groups, revision plans.
Inbox	Enquiries, bookings, orders and notifications.	Conversations, support tickets, message requests.
Profile	Account, campus membership, favorites, settings.	Privacy, verification, payment history.

Public website sitemap

- Home: clear housing search, value proposition, trust process and app download links.
- Schools directory: institution → campus → neighborhood → relevant housing and vendors.
- Housing: category/search results → full listing details → enquiry/inspection booking.
- Campus services: vendors, menus, student business listings and useful local guides.
- Student resources: FAQ, campus guides, academic articles, policies and support.
- Become a landlord/vendor: onboarding, plans, requirements and business sign-in.

Business and admin screens

- Property dashboard: inventory, media, price lines, inspection calendar, leads, renewals and reports.
- Vendor dashboard: store profile, catalog, hours, orders, refunds, settlement and customer support.
- Admin workspace: review queue, school configuration, incident management, transaction oversight, analytics and permissions.

Design system direction

Clean, modern and student-friendly. Use navy for trust, teal for primary actions, spacious white surfaces, legible labels and explicit status colors. Maintain large tap targets, accessible contrast, local currency formatting and a restrained animation budget.

CHAPTER 11 / PRODUCT BLUEPRINT

Monetization and financial hypotheses

Keep student search free. Charge for verifiable business value.

Revenue streams - proposed, not validated

STREAM	POSSIBLE MODEL	PRIORITY
Property subscriptions	Example: ₦2,000/month per paying property provider; free limited trial.	First test
Premium property placement	Clearly labeled promotion with no effect on verification.	Later
Food transactions	Example: 5% take rate or fixed vendor fee; check unit economics.	After orders
Premium business profiles	Example: ₦3,000/month for nearby campus services.	Later
Event ticket fees	Small fee only where ticketing and refund ops are ready.	Later
Student subscriptions	Optional advanced study tools; core features remain free.	Optional

Illustrative monthly gross contribution scenario

COMPONENT	ASSUMPTION	AMOUNT
Housing plans	40 providers × ₦2,000/month	₦80,000
Premium vendors	20 vendors × ₦3,000/month	₦60,000
Food fee contribution	1,000 completed orders × ₦100 platform fee	₦100,000
Illustrative total	**Before costs, refunds and tax**	**₦240,000**

Not a forecast. These inputs are an arithmetic scenario, not evidence of adoption. Net margins depend on payment charges, acquisition, verification, hosting, vendor support, refunds, staffing and compliance.

Questions to validate first

- Do landlords prefer a subscription, pay-per-qualified-lead, or payment per actual rented room?
- What percentage of listings remain truly available during peak housing season?
- What does a completed inspection cost to support and verify?
- Can a small food order support payment processing, packaging, delivery and customer support?
- What revenue does a typical campus vendor generate from being visible on the platform?

Student fee policy

Do not place essential details or genuine contact options behind unexpected student paywalls. If student fees are later tested, they must purchase clear optional value, such as a premium convenience service, not basic access to accommodation.

CHAPTER 12 / PRODUCT BLUEPRINT

Launch strategy: one campus, high reliability

Prioritize depth of supply and student trust over geographic breadth.

Campus selection scorecard

CRITERION	WHY IT MATTERS	WEIGHT*
Housing pain and supply gap	There must be an urgent, recurring problem.	30%
Ability to verify in person	Fraud prevention and genuine photos matter.	25%
Founder / team local access	Faster vendor onboarding and support.	20%
Dense student neighborhoods	Better discovery and later food economics.	15%
Student ambassador access	Distribution and feedback loops.	10%

* Suggested decision weights; use actual campus interviews to assess candidates. Example institutions to explore include UNIOSUN (specific campus), OAU, UI or a nearby polytechnic. Do not select a school solely from a generic ranking.

Pilot targets

30–50
Verified available rooms

10–15
Participating vendors

200–300
Students in pilot

Go-to-market channels

- Partner with student representatives, recognized societies and credible campus creators.
- Publish searchable school and neighborhood guides with honest pricing examples.
- Recruit landlords in person; provide help photographing and structuring their listings.
- Use a QR-code and WhatsApp referral workflow that takes students directly to the relevant campus page.
- Run narrow promotions tied to the academic calendar and accommodation search season.

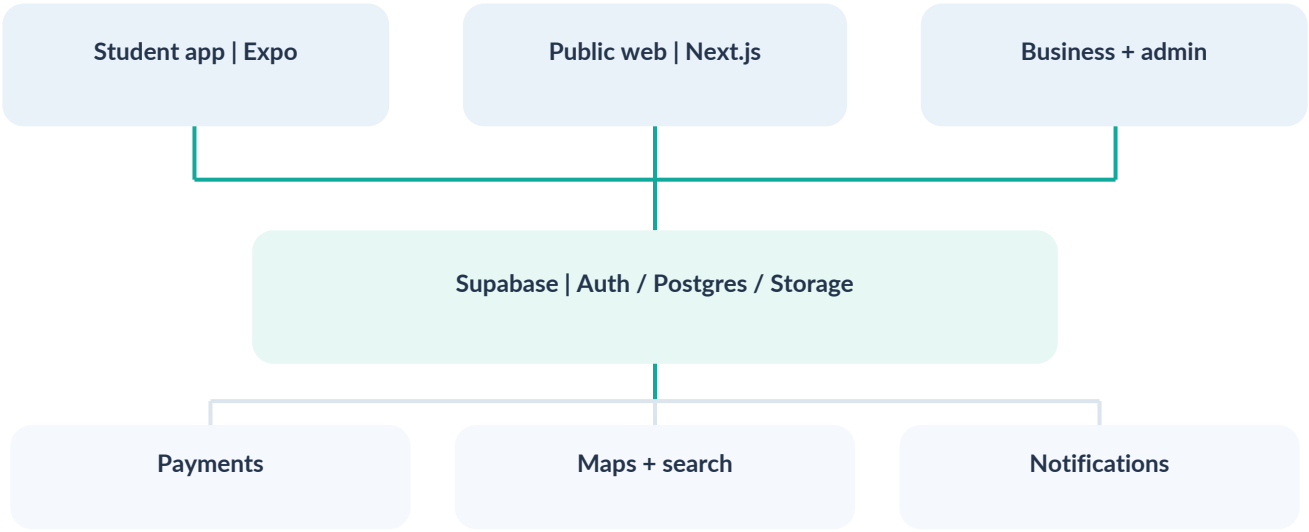
Pilot graduation conditions

Expand only when current listings are repeatedly reconfirmed, qualified student enquiries convert into actual viewings, reports are handled promptly, and businesses demonstrate willingness to pay. Avoid expanding based only on total registrations.

CHAPTER 13 / PRODUCT BLUEPRINT

Technical architecture

A shared stack suited to Expo, Next.js and Supabase.



Recommended components

LAYER	CHOICE	RESPONSIBILITY
Android/iOS app	React Native + Expo	Student experience, push, offline caches.
Web / portals	Next.js + TypeScript	SEO pages, provider and admin dashboards.
Backend	Supabase PostgreSQL / Auth / Storage	Accounts, schools, listings, role-based access and media.
Secure logic	Supabase Edge Functions or server API	Webhook handling, gated writes, notifications, trusted actions.
Location	Map provider + geocoding	Maps, campus coordinates and local search.
Payments	Paystack or verified compatible provider	Plans and qualifying checkout; backend verification.
Monitoring	Error tracking and analytics	Failures, funnels, order and verification metrics.

Engineering conventions

- Monorepo recommended: apps/mobile, apps/web, packages/ui, packages/types, packages/config.
- API contracts and shared validation schemas to minimize app-web drift.
- Use Postgres Row Level Security and role-based authorization on every sensitive table.

- Use background jobs / scheduled tasks to expire stale inventory, send reminders and reconcile webhooks.
- Search/index by campus, region, availability, price and location; optimize images and paginate everything.
- Cache low-risk academic tools and recently viewed non-sensitive content for weak network conditions.

AI adoption rule

Build deterministic features first. Introduce paid OCR or language models only when there is a demonstrated use case, clear usage limits, consent for uploaded data, and measurable improvement over simpler matching and reminders.

CHAPTER 14 / PRODUCT BLUEPRINT

Data model and API boundaries

Key entities and the relationships the development team should preserve.

Core database domains

DOMAIN	TABLES / ENTITIES	IMPORTANT RELATION
Identity / access	profiles, roles, campus_memberships, verification_requests	Member can have a scoped school/campus; roles independently assigned.
Institutions	institutions, campuses, campus_areas, departments, programs	A university may own many campuses and area sets.
Housing	provider_profiles, properties, property_units, amenities, listing_media	Property → many units; unit has own status and pricing.
Housing operations	inspection_requests, availability_checks, verification_audits	Immutable action history where appropriate.
Commerce	vendors, store_items, orders, order_items, fulfillment_events	Orders belong to one vendor unless multi-vendor is designed.
Marketplace	listings, service_requests, conversations, reports	Moderated posts with expiry and ownership.
Academic	courses, terms, timetable_entries, tasks, grade_scales, grades	Grades owned by student and school-specific scale.
Campus life	announcements, events, clubs, event_signups	Source and institutional scope required.
Finance / ops	payment_attempts, transactions, webhook_events, disputes, audit_logs	Idempotency and immutable accounting events.

Non-obvious data requirements

- Keep campus IDs on scoped records; do not assume all branches of an institution share one neighborhood.
- Represent multiple rooms/units beneath one property; avoid duplicating one building for every bed.
- Store rent components and currency separately to calculate honest total move-in costs.
- Track availability_checked_at and verification_checked_at independently.

- Store approximate public location separately from precise inspection-only addresses.
- Preserve source URL, provenance, review status and updated timestamp on campus announcements.
- Retain a transaction ledger and webhook event IDs for reconciliation and idempotency.

API / event examples

ACTION	SERVER ENFORCEMENT
POST inspection request	Authenticate requester; check unit availability; throttle duplicate requests.
PATCH property unit	Check owner role; append audit; prevent unauthorized verified status.
POST order checkout	Recalculate server-side price and vendor stock; create payment intent.
POST payment webhook	Verify signature, idempotency, amount, currency and allowed state change.
REPORT listing	Rate-limit, queue moderation and protect reporter identity.

CHAPTER 15 / PRODUCT BLUEPRINT

Trust, safety, privacy and compliance

The product must reduce rather than amplify campus fraud.

Risk controls

RISK	PREVENTION	RESPONSE
Fake hostel / impersonated landlord	Identity and authority checks; recent media; limited unverified exposure.	Freeze listing, preserve evidence, report/escalate.
Hidden fees	Required structured price fields; full cost display.	Correct/remove misleading entry.
Stale availability	Provider reconfirmation with expiry; student stale-listing reports.	Temporarily unpublish until confirmed.
Harassment or unsafe roommate matching	Opt-in discovery; privacy choices; block/report.	Suspend repeat offenders and escalate serious threats.
Data exposure	RLS, encrypted transport, limited access and retention.	Incident response and applicable notifications.
Food order failure	Vendor acceptance, order tracking, defined refunds.	Issue support and reconcile transaction.

Privacy design

- Publish a clear privacy notice explaining collection, purposes, sharing, retention and rights.
- Collect minimum identity/education data; avoid raw student IDs unless necessary for a justified verification use case.
- Keep government IDs and landlord evidence in restricted storage, with limited reviewers and scheduled deletion.

- Offer account deletion, correction and export pathways appropriate to applicable law.
- Restrict marketing messages and location data use according to user choice and legitimate purpose.
- Handle minors cautiously: not all students are adults; assess applicable age/consent requirements.

Nigeria's Data Protection Act, 2023 is a key legal baseline. The detailed legal approach, business registration, consumer protections, property/agency law and any payment-services obligations require professional review before launch.

Payment and property boundaries

A payment gateway offering split settlements does not automatically provide escrow, renter protection, or authority to hold third-party funds. Prioritize direct marketplace subscriptions first; design rental deposits and any wallet-like model only with competent legal counsel and a suitable regulated provider.

CHAPTER 16 / PRODUCT BLUEPRINT

Operations, service quality and human workflows

A small launch needs specific owners for verification, support and updates.

Operating playbooks

QUEUE	CHECKS	PROPOSED TARGET*
New property submission	Proof of authorization, owner contact, media, location, price completeness.	Initial review within 2 business days
Availability reconfirmation	Reminder, owner response, random spot check.	Every 7-14 days during peak season
Student inspection request	Provider notification, acceptance, no-show/cancellation tracking.	Response within 24 hours
Fraud / safety report	Severity review, temporary pause where warranted, evidence log.	High-risk triage same day
Food order issue	Order events, vendor communication, refund eligibility.	Respond within staffed operating hours

* Internal proposed service objectives, not promised service levels. Revise based on staffing and demand.

Minimum team roles

- Founder / developer: product, code, infrastructure and analytics.
- Campus operations lead or trusted local contractor: listing checks, provider onboarding, issue escalation.
- Part-time support/moderation coverage: user reports, scheduling and common questions.
- Campus ambassadors: research, referral education and content - never independent verification approvals.

Quality cadence

- Daily: inspect high-risk flags, urgent messages, failed payments and broken links.
- Weekly: sample live housing listings, review drop-off funnels, contact inactive providers.

- Monthly: unit economics, student satisfaction, conversion quality, refunds and retention.

CHAPTER 17 / PRODUCT BLUEPRINT

Phased delivery roadmap

A practical order of execution, with gate reviews between stages.

STAGE	BUILD / LEARN	EXIT GATE
Phase 0 | Validation	Interviews, campus selection, waitlist, supply inventory, brand and UX prototype.	Real user workflows validated; local verification feasible.
Phase 1 | Housing MVP	Auth/campus, SEO housing website, listings, filters, verification, inspections, landlord/admin tools, basic study planner.	Reliable room supply; inspection usage; manageable incident rate.
Phase 2 | Commerce	Vendor directory, pickup ordering, secondhand goods, service profiles.	Fulfillment and unit economics proven for one zone.
Phase 3 | Academic / community	Course planner, CGPA, verified notices, clubs, events, learning resources.	Recurring student engagement and moderation capacity.
Phase 4 | Scale	Other schools, delivery expansion, advanced AI, opportunities, partnerships.	Repeatable playbook and healthy campus contribution margin.

Illustrative sequencing, not a fixed deadline

WINDOW	FOCUS
Weeks 1-2	Research, institution choice, product scope and supply partner conversations.
Weeks 3-4	UX, data structure, verification policy and early provider collection.
Weeks 5-8	Implement housing MVP, public web, student app, provider/admin workflow.
Weeks 9-10	Functional QA, security review, seed real listings and closed beta.
Weeks 11-12	Pilot launch, observe inspection funnel, fix issues and reassess pricing.

The 12-week outline is a planning illustration for focused execution, not a delivery commitment. Verification access, app store reviews, team capacity and payment/compliance work may change the schedule.

Avoid premature build scope

> **MVP discipline**
> Do not build AI tutoring, nationwide multi-vendor fulfillment, rent financing, school data scraping, a wallet, or a driver fleet into the first release. These raise complexity without proving that the core housing marketplace works.

CHAPTER 18 / PRODUCT BLUEPRINT

MVP requirements and acceptance criteria

A build-ready product requirement checklist for the first public release.

Priority-0 epics

EPIC	ACCEPTANCE CHECK
Institution setup	Admin can add schools, campuses and areas; student search never mixes campus inventory incorrectly.
Accounts and roles	A student can sign up; provider cannot edit another provider's property; protected files remain private.
Housing catalog	Public listing has room type, status, honest price breakdown, amenities, media and contact/inspection action.
Search	Student can filter by school, campus, type, total rent range and availability; zero-result state is useful.
Verification	No provider can grant itself the verified badge; admin approval stores date, scope and evidence reference.
Inspection requests	Student submits one request, owner responds, status history is visible, cancel/reschedule works.
Landlord portal	Owner can add/edit/pause rooms, respond to leads, and confirm current availability.
Admin operations	Admin can review, hide, reject, suspend, resolve and audit major actions.
Academic basics	Student can add timetable entries and deadline reminders; data survives logout/reinstall.
Feedback and reports	Student can flag misleading listing and receive acknowledgment without exposing identity publicly.

Release quality checks

- Critical flows tested on low-end Android hardware and slow/mobile networks.
- Broken media, empty search, invalid money fields and offline states handled gracefully.
- RLS security tests cover student, provider, moderator and anonymous access.
- Server-side checks protect pricing, booking state and verification permissions.
- Accessibility basics: visible focus, readable text, contrast and supported screen readers.
- Analytics events verify that search, save, enquire, inspection acceptance and reports work.

CHAPTER 19 / PRODUCT BLUEPRINT

Success metrics and experiment plan

Measure delivered student value, not vanity downloads.

North-star candidate

> **Qualified accommodation matches per active campus**
> Count student housing enquiries that reach an acknowledged, available property and proceed to a meaningful inspection or directly verified follow-up. Confirm the exact event definition during pilot.

STAGE	MEASURES	WHY IT MATTERS
Supply	Active verified units, listing freshness, availability confirmation rate.	Prevent empty / stale inventory.
Discovery	Search-to-listing click, filter use, zero-result searches.	Assess relevance and matching.
Trust	Reports per 100 listings, fraud incidents, verified-provider share.	Safety and reputation.
Conversion	Enquiry-to-reply, inspection booking rate, completed viewing rate.	Actual housing outcomes.
Retention	Weekly active students using non-housing tools.	Long-term campus utility.
Monetization	Paid provider conversion, renewal, gross and net contribution.	Sustainable business.

Pilot experiments

- **Listing transparency:** compare full fee breakdown against basic rent-only layouts; measure enquiries and complaints.
- **Supply monetization:** interview and test free trial → subscription versus paid qualified leads.
- **Onboarding:** compare immediate account requirement versus browse-first flow.
- **Repeat engagement:** measure whether timetable reminders encourage voluntary weekly returns.
- **Campus marketing:** compare ambassadors, SEO guides and vendor QR codes by qualified activation.

CHAPTER 20 / PRODUCT BLUEPRINT

Risk register and decision log

What can derail the idea, and how to limit exposure.

RISK	LEVEL	MITIGATION
Fake, stale or unauthorized housing	High	Verification ladder, recurring checks, reports and fast takedown.

RISK	LEVEL	MITIGATION
Insufficient local inventory	High	Supply-first campus onboarding, targeted landlord outreach.
No willingness to pay	High	Pricing interviews and trials before fixed plans.
Payment disputes / legal exposure	High	Avoid rent holding, document refund policies, legal review.
Vendor food failures	Medium	Begin with pickup, small vendor cohort, clear order states.
Scope creep	High	Stage gates and strict P0/P1/P2 backlog.
School rumors / misinformation	Medium	Source labels, moderation and verified publisher process.
Cost of operations	Medium	One campus, repeatable checklists, unit economics monitoring.

Important open decisions

- Final brand name, domain strategy and tone of voice. CampusOS is a working name only.
- First institution and campus, after direct research and ability to verify providers.
- Subscription versus lead-based landlord pricing and free-trial duration.
- Whether inspections are coordinated by the platform or directly by providers.
- Whether the launch starts as mobile web plus app or web-first with an app soon after.
- Which academic basics provide enough ongoing utility without distracting from housing.

CHAPTER 21 / PRODUCT BLUEPRINT

The first 30 actions

A realistic founder checklist for moving from blueprint to implementation.

#	WORKSTREAM	ACTION
01	DISCOVERY	Select 2-3 candidate campuses and create a weighted comparison.
02	DISCOVERY	Interview 10 newly admitted students and capture their last housing search.
03	DISCOVERY	Interview 10 returning students about total fees, fraud and renewals.
04	DISCOVERY	Interview 10 students about food, academics and daily services.
05	DISCOVERY	Interview 10 landlords / agents about vacancies and pricing.
06	DISCOVERY	Interview 10 vendors about orders, margins and support.
07	DISCOVERY	Map campus neighborhoods, school gates and common hostels.
08	DISCOVERY	Draft a verified listing checklist and disclosure standard.
09	DISCOVERY	Create an early student waitlist and landlord interest form.

#	WORKSTREAM	ACTION
10	STRATEGY	Choose the launch campus and confirm a local operations path.

#	WORKSTREAM	ACTION
11	STRATEGY	Choose housing-first external positioning and core brand system.
12	STRATEGY	Write MVP success measures and explicitly exclude Phase 2+ scope.
13	DESIGN	Map student housing discovery to physical inspection flow.
14	DESIGN	Map landlord onboarding, verification and inventory management.
15	DESIGN	Sketch app home, search, property, inspection and saved rooms.
16	DESIGN	Sketch public SEO pages and business/admin dashboards.
17	DESIGN	Test screens with five students and adjust language / fee clarity.
18	ENGINEERING	Create monorepo, environments, type checks and CI pipeline.
19	ENGINEERING	Define institution/campus and role-based database schema.
20	ENGINEERING	Implement auth, sessions, RLS and private document storage.

#	WORKSTREAM	ACTION
21	ENGINEERING	Build housing CRUD, unit availability and verification auditing.
22	ENGINEERING	Build responsive discovery/search/property detail pages.
23	ENGINEERING	Build student app browse, save and inspection requests.
24	ENGINEERING	Build owner dashboard and admin review queue.
25	ENGINEERING	Add reminders, notifications and basic academic planner.
26	QUALITY	Test permissions, fraud reports, spam throttling and offline behavior.
27	QUALITY	Seed real, permissioned listings and verify availability in person.
28	LAUNCH	Recruit 200-300 pilot students using school-relevant channels.
29	LAUNCH	Review weekly funnels, support cases and provider value.
30	LAUNCH	Decide whether to improve housing or unlock commerce features.

Decision rule for moving forward

A better outcome is a useful, operational housing service at one campus - with documented inspections, honest listings and repeat users - than a polished nationwide app with empty modules.

CHAPTER 22 / PRODUCT BLUEPRINT

References, assumptions and ownership

Sources inform the need and constraints; they do not replace campus-specific discovery.

Public sources for background and implementation

Housing market context

The Guardian, "How demand spike, inflation worsen Nigeria's student housing crisis" (2026).

https://guardian.ng/property/how-demand-spike-inflation-worsen-nigerias-student-housing-crisis/

Student experience / pricing context

Sunday PUNCH, "Shylock landlords, greedy agents choke off-campus students with high rents" (2026).

https://punchng.com/shylock-landlords-greedy-agents-choke-off-campus-students-with-high-rents/

Data protection law

National Cybersecurity Coordination Centre, Nigeria Data Protection Act 2023 resource.

https://nccc.gov.ng/resources/nigeria-data-protection-act-2023

Payment capability

Paystack Developer Documentation, Split Payments.

https://paystack.com/docs/payments/split-payments/

Official funding information

Nigerian Education Loan Fund official portal.

https://nelf.gov.ng/

Assumptions requiring validation

- Sample pricing (₦2,000/month housing provider, ₦3,000/month premium vendor, 5% food fee) is proposed, not tested.
- Pilot targets and 12-week sequencing are planning assumptions, not forecasts or guaranteed delivery dates.
- CampusOS is a working product name; naming rights, domain availability and trademark clearance are pending.
- Legal and payment decisions need specialist review before launching regulated or sensitive payment flows.
- All described images, maps and content require appropriate permission and provenance.

Ownership and next output

Use this blueprint to create: (1) a validated campus launch brief, (2) screen-by-screen UI specification, (3) engineering backlog with ticket IDs and test cases, (4) Supabase SQL migrations and access policies, and (5) an operational verification manual. Those deliverables should follow the first round of real campus research.